

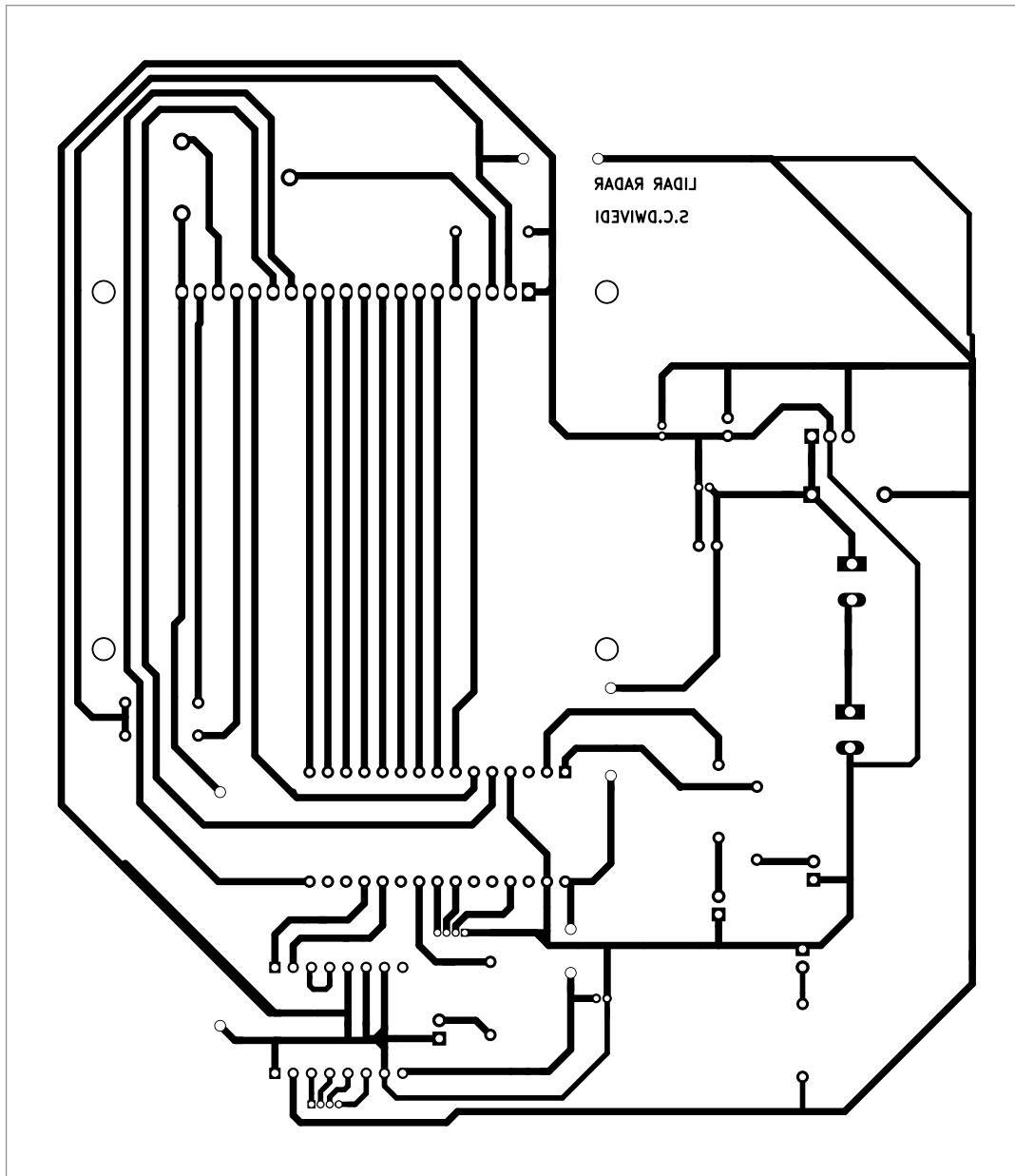


Fig. 4: Actual-size solder-side PCB layout

The GLCD screen used here has 128×64 -pixel resolution and is divided into two equal halves of 64×64 pixels. By changing CS1 (pin 15) and CS2 (pin 16) pins of GLCD, the left and right parts of the screen are activated to display the data. The data and command are of 8-bit length (one byte) and differentiated by selecting status of RS pin. Preset VR1 is used to adjust GLCD screen's contrast.

The lidar module is fitted on

the shaft of a stepper motor (see Fig. 2). The selected stepper motor has small NEMA17 frame size with 1.8-degree step and 1A coil current

rating, or as per individual choice. The selected stepper motor driver module A4988 can deliver up to 2A current to the stepper motor.

The brain of the system is compact, bread-board-compatible Arduino Nano board, which uses surface mount ATmega328 as its microcontroller. The Arduino Nano board can be placed on the PCB or bread-board using Berg strips quite comfortably.

Setup and usage

Upload the sketch to Arduino Nano and connect all the components as shown in the circuit diagram. Attach the lidar module to the stepper motor shaft with the lidar sensor facing towards the object. The actual-size solder-side

PCB layout of obstacle detection using Lidar module is shown in Fig. 4 and its component layout in Fig. 5.

When power supply to the circuit is switched on through switch S1, the display shows 'LIDAR RADAR' and the stepper motor rotates to its starting position. Then the stepper motor shaft rotates right and left (within 60 degrees) along with the lidar module attached to it to detect the object.

On detection, the distance of the

EFY Note

The source code of this project is available for free download at www.electronicshub.in